

**From Person-Specific Dynamics to Population Inference: A Computationally Efficient
Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel
Dynamic Models**

Ivan Jacob Agaloos Pesigan¹, Michael A. Russell^{1, 2}, and Sy-Miin Chow³

¹Edna Bennett Pierce Prevention Research Center, The Pennsylvania State University

²Department of Biobehavioral Health, The Pennsylvania State University

³Department of Human Development and Family Studies, The Pennsylvania State University

Author Note

Ivan Jacob Agaloos Pesigan  <https://orcid.org/0000-0003-4818-8420>; Michael A. Russell  <https://orcid.org/0000-0002-3956-604X>; Sy-Miin Chow  <https://orcid.org/0000-0003-1938-027X>.

This research was supported by the Prevention and Methodology Training Program (PAMT), through a T32 training grant from the National Institute on Drug Abuse (NIDA; T32 DA017629; MPIs: J. Maggs and S. Lanza), as well as by grant UL1TR002014-06 from the National Center for Advancing Translational Sciences (NCATS).

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer using SLURM for job scheduling (Yoo et al., 2003), GNU Parallel to run the simulations in parallel (Tange, 2021), and Apptainer to ensure a reproducible software stack (Kurtzer et al., 2017, 2021).

Correspondence concerning this article should be addressed to Ivan Jacob Agaloos Pesigan, Edna Bennett Pierce Prevention Research Center, College of Health and Human Development, The Pennsylvania State University, 320 Biobehavioral Health Building, University Park, PA 16802 or by email (ijapesigan@psu.edu).

Links

Research Compendium

The data and materials for this study are available on OSF (<https://osf.io/uenr7>) and GitHub (<https://github.com/jeksterslab/manMetaVAR>, <https://jeksterslab.github.io/manMetaVAR/index.html>).

fitVARMxID R Package

Source code and documentation for the `fitVARMxID` R package are available on GitHub (<https://github.com/jeksterslab/fitVARMxID>, <https://jeksterslab.github.io/fitVARMxID/index.html>).

metaDyn R Package

Source code and documentation for the `metaDyn` R package are available on GitHub (<https://github.com/jeksterslab/metaDyn>, <https://jeksterslab.github.io/metaDyn/index.html>).

Single Replication from the Simulation Study

This vignette presents a complete walk-through of one representative Monte Carlo replication, including data generation, estimation for each competing method, and selected output and diagnostics. Convergence diagnostics for a representative BMLVAR (Bayesian multilevel vector autoregressive model) replication, including PSR, ESS, and trace plots, are provided via the `Mplus` output section of the simulation-replication vignette.

<https://jeksterslab.github.io/manMetaVAR/articles/sim-rep.html>

Empirical Illustration: Meta-Analysis of Affect Dynamics

This vignette reproduces the empirical example reported in the manuscript, including data preprocessing, person-specific VAR estimation, second-stage meta-analytic synthesis, and the resulting population-level affect-dynamics summaries.

<https://jeksterslab.github.io/manMetaVAR/articles/empirical.html>

Benchmarking MetaVAR Against an Uncertainty-Uncorrected Two-Stage Approach and Bayesian Multilevel VAR

This vignette reports the benchmarking analysis comparing the computational demands of MetaVAR, the uncertainty-uncorrected two-stage benchmark, and BMLVAR under a common simulation setting.

<https://jeksterslab.github.io/manMetaVAR/articles/benchmark.html>

Containers for Reproducibility

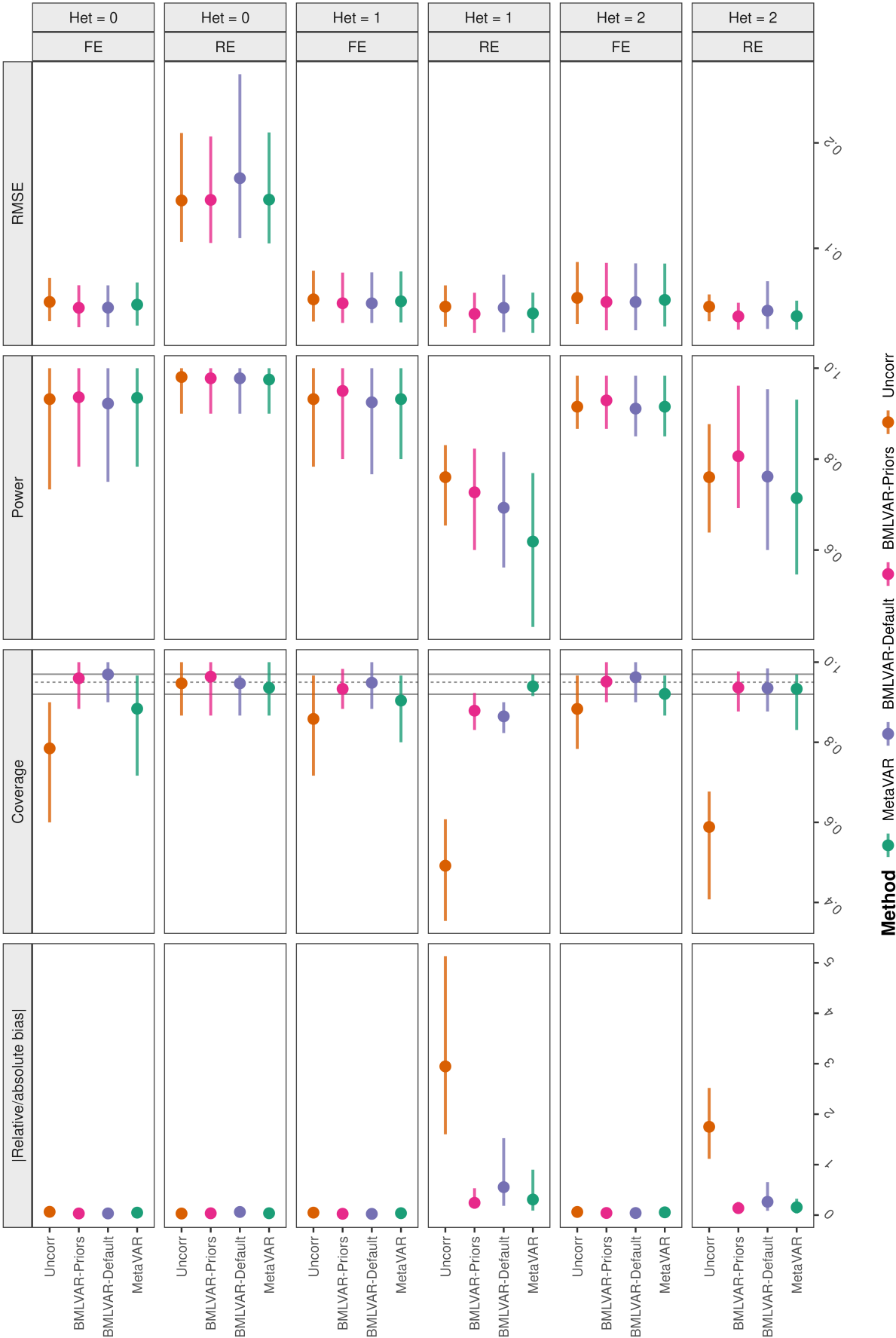
This vignette provides the containerized computing environments and step-by-step instructions needed to reproduce the analyses with the required software stack and dependencies.

<https://jeksterslab.github.io/manMetaVAR/articles/containers.html>

Figure 1
Overall Performance

Overall method performance across simulation design cells and heterogeneity conditions

Points are means across design cells; horizontal lines show the min-max range. Coverage panels include nominal .95 and Bradley's liberal .92-.97 band.



Coverage heatmap by parameter, design cell, and method

Negative values indicate undercoverage; zero indicates nominal coverage.

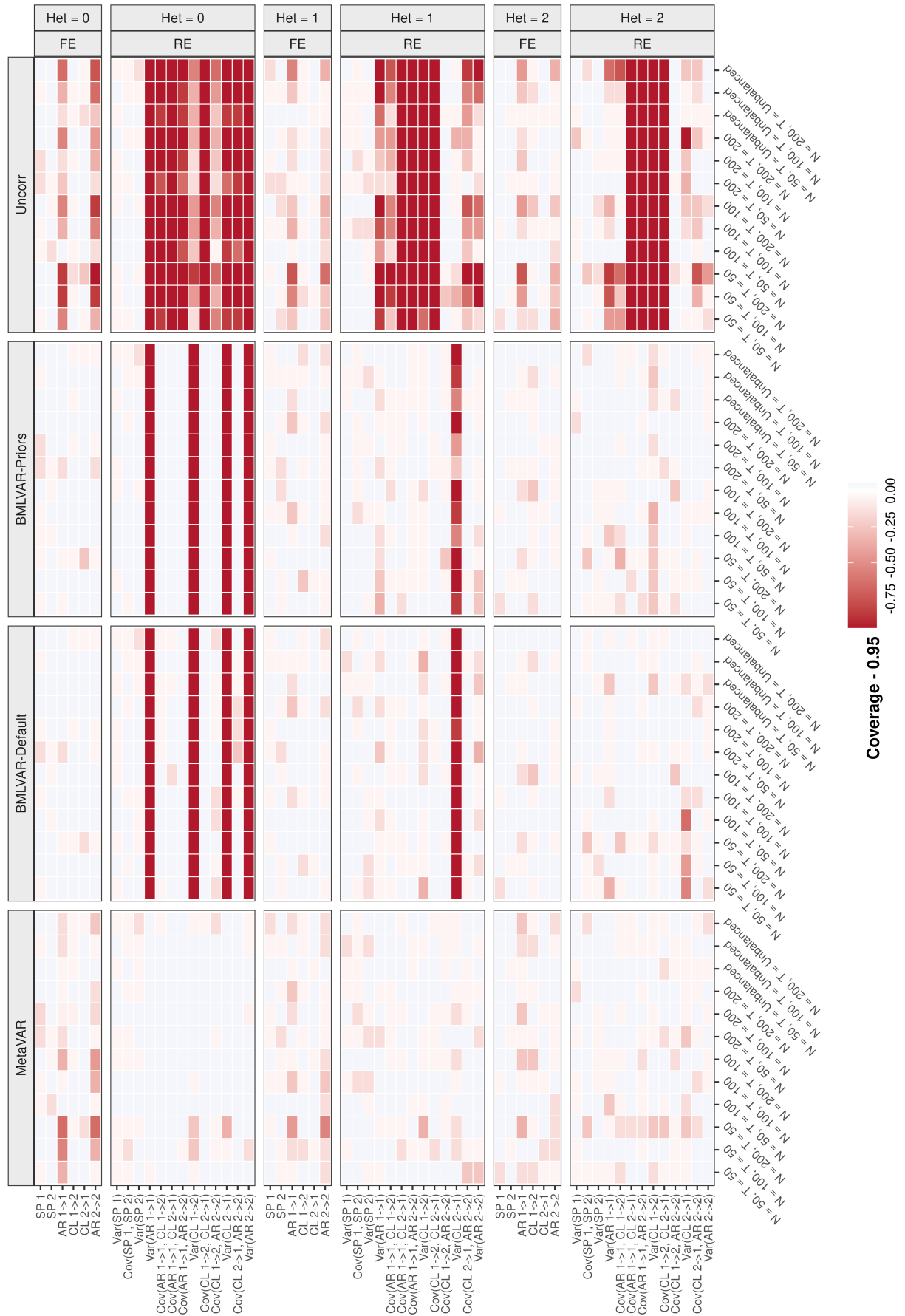
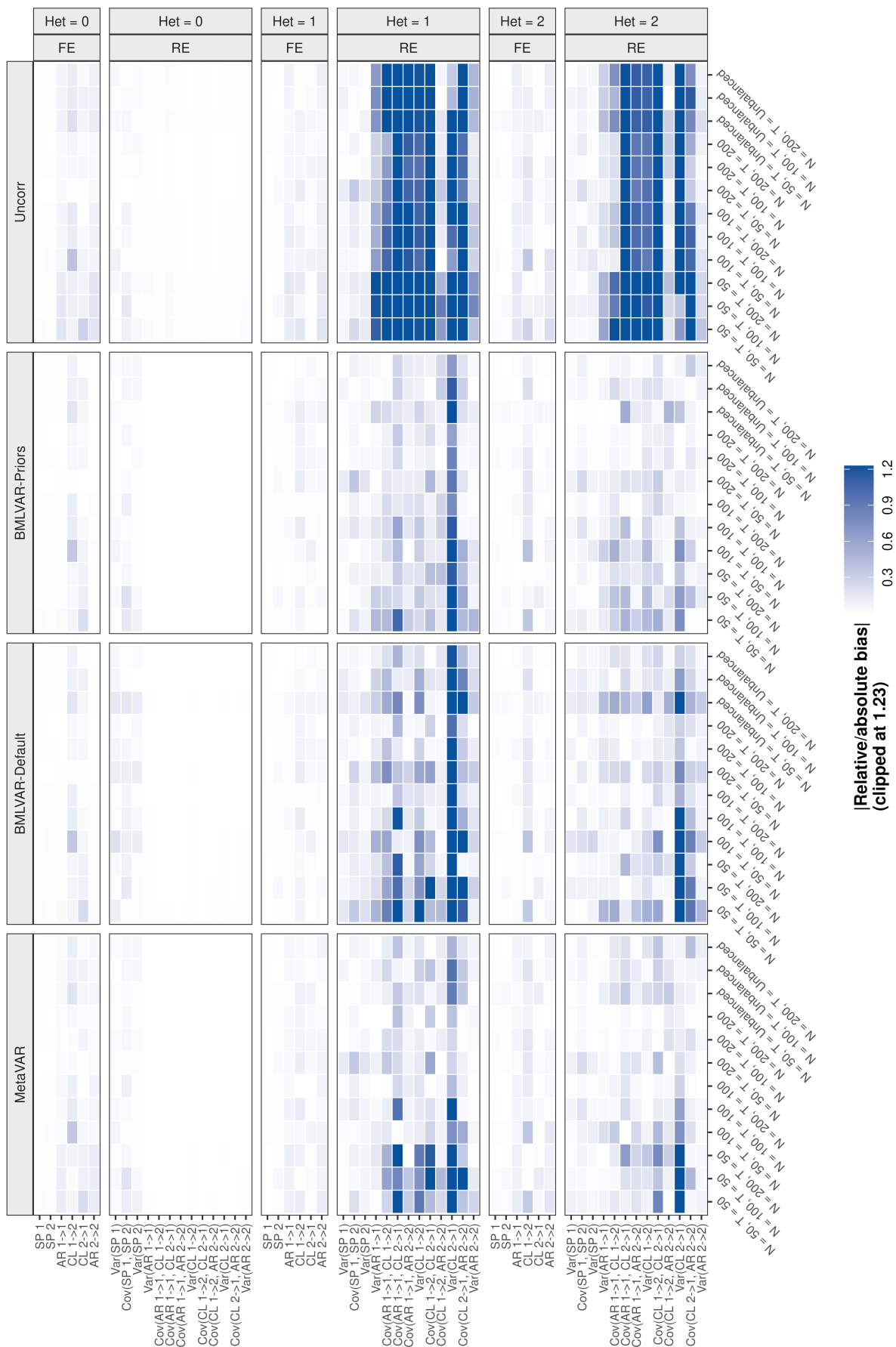


Figure 3
Relative Bias

Bias-magnitude heatmap by parameter, design cell, and method

To keep the heatmap readable, the bias magnitude is clipped at the 95th percentile.

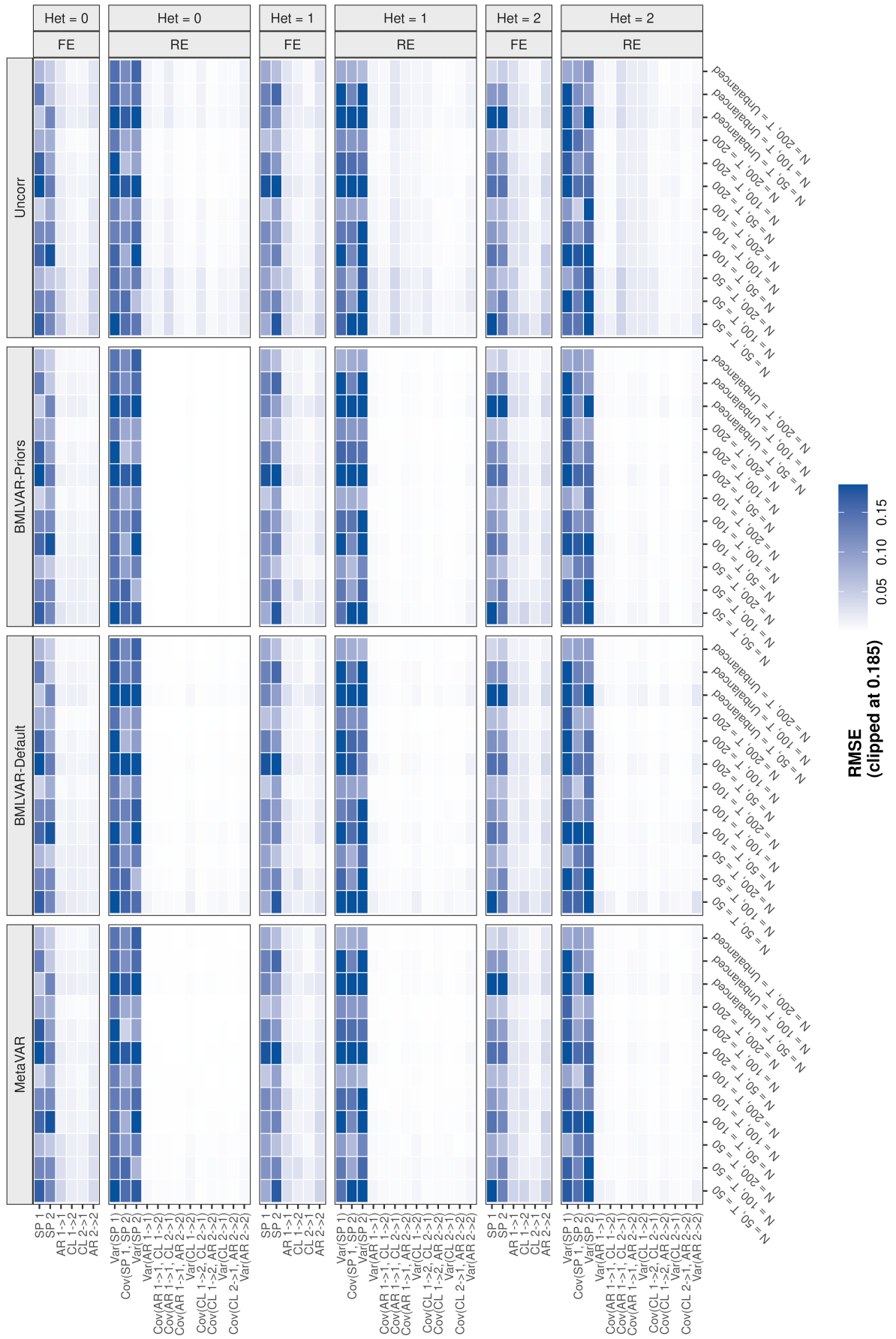


For nonzero parameters, cells show absolute relative bias; for zero parameters, cells show absolute bias. The heatmap shows magnitude only, not direction.

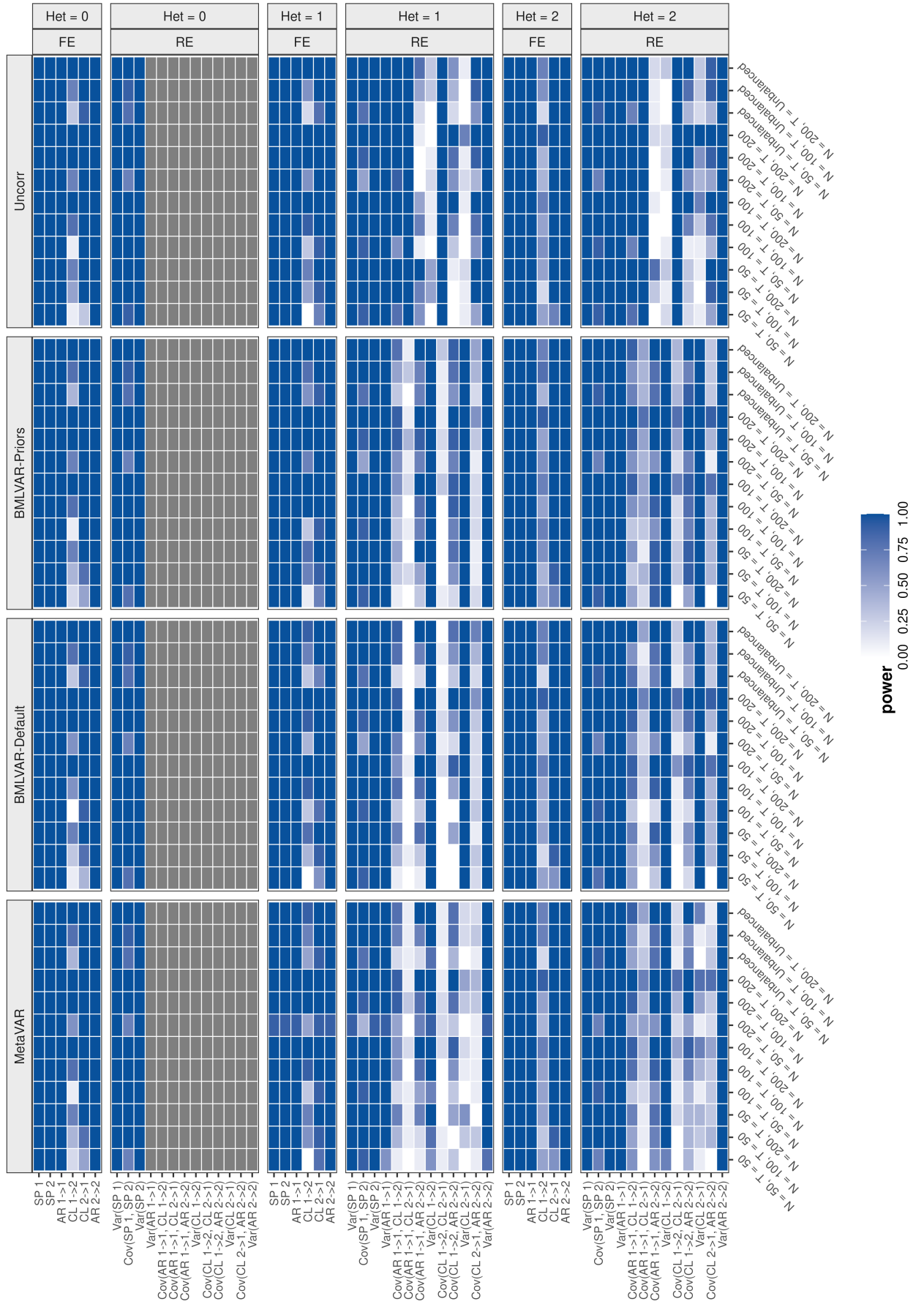
Figure 4
RMSE

RMSE heatmap by parameter, design cell, and method

To keep the heatmap readable, RMSE is clipped at the 95th percentile.



Power heatmap by parameter, design cell, and method

Figure 5
Power